

MINHO KIM

 minho.me  mhk@berkeley.edu

 Google Scholar |  ResearchGate |  Twitter | ORCID

RESEARCH INTERESTS

- **Geospatial Analysis:** Remote Sensing, Computer Vision, GIScience, Network Science
- **Machine Learning:** Deep Learning, Geospatial AI, Explainable AI
- **Environmental Planning:** Risk & Resilience, Natural Hazards, Sustainable Development

EDUCATION

-  **University of California, Berkeley** **Sep 2021 – May 2026**
Ph.D. Environmental Planning
Dissertation: Data-Driven Planning for Natural Hazard Risk Management
Advisors: Marta C. González, John Radke
Exam Committee: John Radke, Marta C. González, Iryna Dronova, Solomon Hsiang
-  **Seoul National University** **Mar 2017 – Feb 2021**
M.S. Civil & Environmental Engineering
Thesis: Local Climate Zone Classification Using Multi-Scale Convolutional Networks
Advisor: Yongil Kim
-  **Seoul National University** **Sep 2012 – Feb 2017**
B.S. Civil & Environmental Engineering
Thesis: Monitoring North Korea's 4th Nuclear Test Site with Sentinel-1A Data Using DInSAR
Advisor: Yongil Kim

RESEARCH EXPERIENCE

-  **Graduate Student Researcher (HuMNet Lab)** **Jan 2022 – Present**
Advisor: Marta C. González *UC Berkeley*
 - Mapping high resolution fuels and vegetation using deep learning [C10]
 - Fire spread simulations via multi-objective optimization [C13], [P6] coupled with downscaled wind [C16]
 - Quantified homeowner responsibility of wildfire risk mitigation and simulated community-scale mitigation strategies via network dismantling [C15], [W1]
 - Spatiotemporal analysis of global pyromes under synchronous fire weather risk and human exposure [W5]
-  **Researcher (Center for Catastrophic Risk Management)** **Jan 2025 – Present**
 - Assisted with developing a hybrid agent-based, system dynamics model to investigate wildfire management response and evacuation during the 2025 Los Angeles fires [W3], [W4]
 - Reviewed natural hazard modeling and data sources for catastrophic wildfire events
-  **Visiting Researcher (Catalan Fire Service)** **July 2024 – Sep 2024**
Advisor: Marc Castellnou *Barcelona, Spain*
 - Modeled fire potential polygons and networks using fire spread simulations and hydrology-inspired tools for operational decision support (fire suppression, risk management, tactical maneuvers) [C12], [P7]
 - Improved existing WUI maps using structural separation distance and permeability metrics
-  **Graduate Student Researcher (River Lab)** **May 2023 – June 2024**
Advisors: Mathias Kondolf, John Radke *UC Berkeley*
 - Developed decision-making framework to estimate bulking factors and protect critical infrastructure against compound/cascading hazards (post-fire debris flows) [C14], [W2]
 - Built GUI to aggregate data from APIs and GIS data related to post-fire debris flow probability
-  **Research Assistant (SPINS-RS Lab)** **Mar 2019 – Feb 2021**
Advisor: Yongil Kim *Seoul National University*
 - Urban Remote Sensing: Generated high resolution Local Climate Zone classification maps multi-scale CNNs (~80% accuracy) [C7] and multi-scale, multi-level attention CNNs (~90% accuracy) trained with Sentinel-2 images and multi-modal GIS data [P4].

- Renewable Energy: Predicted photovoltaic power of solar farms ($< 5\%$ Normalized MAE) using large-scale time series of geostationary satellite images and multi-source meteorological data [C2], [C4], [P3].
- Image Fusion: Developed a spatiotemporal image fusion model in Matlab to produce disaggregated Landsat-8 thermal images in heterogeneous urban areas [C6].
- Change Detection/Monitoring: Applied radiometric calibration to help detect and monitor burn scars using change detection results from Sentinel-2 and PlanetScope images [C3], [P2].

Undergraduate Research Assistant (SPINS-RS Lab)

Aug 2016 – Feb 2017

Advisor: Yongil Kim

Seoul National University

- Analyzed ground deformations in inaccessible, remote areas using dInSAR with Sentinel-1 SAR images.
- Carried out fieldwork and experiments using a ground-based hyperspectral imager to monitor crop health.

Research Assistant (Lawson Health Research Institute)

Sep 2011 – Jan 2012

Advisor: Jeffrey Carson

London, Canada

- Photoacoustic image reconstruction of a line source using multiple regularization percentages with maximum intensity projection using Matlab.

WORK EXPERIENCE

Researcher at Institute of Construction & Env. Eng.

Mar 2021 – Aug 2021

Advisor: Yongil Kim

Seoul National University

- Mapped inaccessible areas using deep learning with very high resolution satellite imagery [C8].

PR Manager

Mar 2021 – Aug 2021

Education & Research Program (InfraSPHERE)

Seoul National University

Lab Manager

Mar 2021 – Aug 2021

SPINS-RS Lab

Seoul National University

HONORS & AWARDS

Outstanding Graduate Student Instructor Award	2024
🏆 UC Berkeley (GSI Teaching and Resource Center)	
ICE-KSCE Master's Thesis Award	2021
🏆 Institution of Civil Engineers (UK) & Korean Society of Civil Engineers	
Best Student Paper Award at ISRS2021	2021
🏆 Korean Society of Remote Sensing and Gaia3D	
Environmental Geospatial Data Idea Contest (Excellence Award)	2020
🏆 Ministry of Environment, South Korea	
SPINS Lab (Outstanding Research Award)	2020
🏆 Seoul National University	
Student Competition using Meteorological Satellites (Research Award)	2019
🏆 Korean Meteorological Administration	

SCHOLARSHIPS

AEP Student Scholarship	2025
🏆 Association of Environmental Professionals (AEP)	
Beatrix C. Farrand Memorial Fellowship for Research	2024, 2025
🏆 UC Berkeley (Dept. of Landscape Architecture & Environmental Planning)	
Beatrix C. Farrand Memorial Fellowship for Conference Travel	2023, 2025
🏆 UC Berkeley (Dept. of Landscape Architecture & Environmental Planning)	
Robert N. Colwell Memorial Fellowship	2023
🏆 The American Society for Photogrammetry and Remote Sensing	
Brain Korea 21 Plus Scholarship	2019–2021
🏆 National Research Foundation of Korea	
Merit-based Scholarship	2014–2017, 2019
🏆 Seoul National University	
National Scholarship for Science and Engineering	2013, 2014
🏆 Korea Student Aid Foundation	

PUBLICATIONS

* Equal contribution, † Corresponding author

Preprints & Working Papers

- [W6] **Minho Kim**, Marc Castellnou, Marta C. González. “Unprecedented extreme fire behavior in South Korea.” To be submitted to *Environmental Research Letters*
- [W5] **Minho Kim**, John Radke, Maryam Hosseini, Marta C. González. “Spatiotemporal analysis of global pyromes and synchronous wildfire risk.”
- [W4] Clara Larson, Lian Plass, **Minho Kim**, Louise Comfort. “Proposing a Modeling Approach for Simulating Catastrophic Risk in Complex Adaptive Systems.” To be submitted to *Natural Hazards Review*.
- [W3] Clara Larson, Lian Plass, **Minho Kim**, Louise Comfort. “Modeling Cascading Risks in Complex Adaptive Systems: Case Study of the 2025 Los Angeles Wildfires.” To be submitted to *Risk Analysis*.
- [W2] **Minho Kim**, Adrienne Dodd, John Radke, G. Mathias Kondolf. “Multi-criteria decision-making for cascading hazards: Case study of post-fire debris flows in Northern California.”
- [W1] **Minho Kim**[†], Harrison Raine, John Radke, Marta C. González. “A Network Approach to Mapping Responsibility in Wildfire Risk Mitigation.” Under review in *Landscape and Urban Planning*.
- [PP1] Cristobal Pais, **Minho Kim**, Yanyan Xu, John Radke, Marta C. González. “An interdisciplinary data-science approach to managing natural hazards risk. [arXiv:2407.07270](https://arxiv.org/abs/2407.07270).”

Peer Reviewed Journal Papers

- [P7] **Minho Kim**, Marc Castellnou, Marta C. González. (2025). “Modeling potential fire spread polygons and networks for suppression strategies”, *International Journal of Disaster Risk Reduction*, 130, 105853
- [P6] **Minho Kim**, Cristobal Pais, Marta C. González. (2025). “Fire Spread Simulations Using Cell2Fire on Synthetic and Real Landscapes”, *Scientific Reports*, 15 25173 (2025).
- [P5] Xihan Yao, **Minho Kim**, Iryna Dronova, Joe McBride, G. Mathias Kondolf, John Radke. (2025). “Community-scale microclimate simulation using Airborne Laser Scanning and object-based urban tree classification”, *Landscape and Urban Planning*, 263, 105420.
- [P4] **Minho Kim**, Jeong, D. & Kim, Y. (2021). “Local climate zone classification using a multi-scale, multi-level attention network”, *ISPRS Journal of Photogrammetry and Remote Sensing*, 181, 345-366.
- [P3] **Minho Kim**, Song, H. & Kim, Y. (2020). “Direct short-term forecast of photovoltaic power through a comparative study between COMS and Himawari-8 meteorological satellite images in a deep neural network”, *Remote Sensing*, 12(15), 2357.
- [P2] **Minho Kim**, Jung, M. & Kim, Y. (2019). “Histogram matching of Sentinel-2 spectral information to enhance Planetscope imagery for effective wildfire damage assessment”, *Korean Journal of Remote Sensing*, 35(4), 517-534.
- [P1] Kim, Y., **Minho Kim**, Choi, J. & Kim, Y. (2017). “Image fusion of spectrally nonoverlapping imagery using SPCA and MTF-based filters”, *IEEE Geoscience and Remote Sensing Letters*, 14(12), 2295-2299.

Conference & Workshop Papers

- [C16] **Minho Kim**, & Gonzalez, M.C. (2026). “Coupling Cell2Fire with Downscaled High Resolution Wind Data”, *NHERI Computational Symposium. Berkeley, USA. May 28-29, 2026*.
- [C15] **Minho Kim**, Harrison Raine, John Radke & Gonzalez, M.C. (2025). “A Network Modeling Approach to Quantify Homeowner Responsibility for Wildfire Risk Mitigation in the Wildland Urban Interface”, *AGU 2025. New Orleans, USA. Dec 15-19, 2025*.
- [C14] **Minho Kim**, Dodd, A., Radke J., Ulyashin V., Serra-Llobet, A., Falzone, A. & Kondolf, M. (2025). “Flood after Fires: A Scalable Decision-Support Framework for Post-Fire Debris Flow Risk”, *AGU 2025. New Orleans, USA. Dec 15-19, 2025*.

- [C13] **Minho Kim**, Pais, C., & Gonzalez, M.C. (2025). “Fire spread simulations using Cell2Fire on synthetic and real landscapes”, *International Fire Ecology and Management Congress. Association for Fire Ecology. New Orleans, USA. Dec 2-6, 2025.*
- [C12] **Minho Kim**, Castellnou, M. & Gonzalez, M.C. (2025). “Modeling fire potential networks for suppression strategies”, *Conference of Complexity Sciences 2025 (Complex Systems, Data Analytics, and Human Behavior). Siena, Italy. Sep 2-5, 2025.*
- [C11] Yao, X. & **Minho Kim**. (2023). “A Lidar-based Method for 3D Urban Forest Evaluation and Microclimate Assessment, a Case Study in Portland, Oregon, USA”, *AGU23. American Geophysical Union. Dec 11-25, 2023.*
- [C10] **Minho Kim**, Dronova, I. & Radke, J. (2023). “Semantic Segmentation of Enhanced Landform Maps Using High Resolution Satellite Images”, *IGARSS 2023 IEEE International Geoscience and Remote Sensing Symposium. IEEE. Pasadena, California, US., July 16-21, 2023. (*Session Chair)*
- [C9] Yao, X. & **Minho Kim** (2023). “Exploratory remote sensing data analysis and clustering of urban vegetation and land surface temperature in Portland, Oregon”, *IGARSS 2023 IEEE International Geoscience and Remote Sensing Symposium. IEEE. Pasadena, California, US., July 16-21, 2023.*
- [C8] **Minho Kim**, Kwak, T., Jung, J. & Kim, Y. (2021). “Mapping inaccessible areas using deep learning based semantic segmentation of VHR satellite images with OpenStreetMap data”, *In Proceedings of the 2021 International Symposium of Remote Sensing, Virtual, May 26-28, 2021. (*Awarded Best Student Paper)*
- [C7] **Minho Kim**, Jeong, D., Choi, H. & Kim, Y. (2020). “Developing High Quality Training Samples for Deep Learning Based Local Climate Zone Classification in Korea”, *arXiv preprint, Presented at AI for Earth Sciences Workshop at NeurIPS 2020, Virtual, arXiv:2011.01436.*
- [C6] **Minho Kim**, Cho, K., Kim, H. & Kim, Y. (2020). “Fusion of High Resolution Land Surface Temperature Using Thermal Sharpened Images from Regression-based Urban Indices”, *ISPRS Annals of the Photogrammetry, Remote Sensing and Spatial Information Sciences, 3, (pp247-254).*
- [C5] Song, A., Kim, C., **Minho Kim** & Kim, Y. (2019). “Analysis of Geospatial Technology for Smart City Development: Case Study of South Korea”, *In Proceedings of The 1st Tunisian Smart Cities Symposium, Tunisia, 2019.*
- [C4] Kim, G., Song, H., Kim, **Minho Kim** & Kim, Y. (2019). “Multimodal Merging of Satellite Imagery with Meteorological and Power Plant Data in Deep Convolutional Neural Network for Short-Term Solar Energy Prediction”, *In Proceedings of the 40th Asian Conference on Remote Sensing, Daejeon, South Korea, Oct 14-18, 2019.*
- [C3] **Minho Kim** & Kim, Y. (2019). “Integration of Sentinel-2 Spectral Information with High Spatial Resolution PlanetScope Imagery for Wildfire Damage Assessment”, *In Proceedings of the 40th Asian Conference on Remote Sensing, Daejeon, South Korea, Oct 14-18, 2019.*
- [C2] Song, H., Kim, G., **Minho Kim** & Kim, Y. (2019). “Short-Term Forecasting of Photovoltaic Power Integrating Multi-Temporal Meteorological Satellite Imagery in Deep Neural Network”, *In 2019 IEEE PES Asia-Pacific Power and Energy Engineering Conference (APPEEC), Macao, (pp1-5).*
- [C1] **Minho Kim** & Kim, Y. (2019). “Monitoring the Catastrophic 2018 Mendocino Complex Wildfire Using the Sentinel Constellation”, *In Proceedings of the 2019 International Symposium of Remote Sensing, Taiwan, April 14-17, 2019.*

INVITED TALKS & PANELS

- [T9] **“Data-driven planning for natural hazard risk management”**
Invited seminar
 Seoul National University **Apr 2026**
- [T8] **“Multiscale mitigation of wildfire risk in the natural and built environments”**
Just Climate Futures Exhibition
 UC Berkeley **Apr 2026**
- [T7] **“Geospatial data science for natural hazard risk analysis”**
Invited lecture for CE153: Resilient Cities by Design
 UC Berkeley **Mar 2026**

[T6]	“Shared responsibility and structure separation distance using network modeling and metrics” <i>Invited seminar for the Catalan Fire Service</i> Forest Actions Reinforcement Group (GRAF), Catalan Government <i>Invited seminar for the Pau Costa Foundation</i> Pau Costa Foundation	July 2024 July 2024
[T5]	“Fire spread modeling and superpixel-based fire suppression networks” <i>Invited seminar for the Meteorology and Air Quality group</i> Wageningen University & Research <i>Invited seminar for the Catalan Fire Service</i> Forest Actions Reinforcement Group (GRAF), Catalan Government	July 2024 July 2024
[T4]	“Data-driven planning and modeling for wildfire research using geospatial data science and network science” <i>Invited seminar for the Ecological Sensing Lab</i> Seoul National University <i>Guest lecture for CP4190 Introduction to Climate Change Planning (Fall 2025)</i> Georgia Institute of Technology <i>Guest lecture for CP4190 Introduction to Climate Change Planning (Spring 2024)</i> Georgia Institute of Technology	Aug 2025 Oct 2025 Feb 2024
[T3]	“Exploring Research in the Environmental Field” <i>Berkeley Environmental Economics and Policy Students</i> UC Berkeley	Oct 2021
[T2]	“Urban Remote Sensing” <i>Guest lecture for graduate course 457.544: Satellite Image Interpretation</i> Seoul National University	Apr 2020
[T1]	“Urban Remote Sensing” <i>Seminar for the Interdisciplinary Program in Landscape Architecture</i> Seoul National University	Jan 2020

PATENTS & SOFTWARE

Song, H., Kim, Y., **Minho Kim**, Kim, K. *Convolutional neural networks for short-term photovoltaic forecast using satellite imagery, meteorological data, and power station data*. Patent, South Korea, 2021.

TEACHING

UC Berkeley

Lead Instructor (Course Link)

- GEOG/LDARCH C188: Geographic Information Systems (*Lead Instructor*) Fall 2022
- Teaching Effectiveness: 6.311/7 from 61/169 students (Dept. Average: 6.230/7)

Graduate Student Instructor

- LDARCH/ESPM C289: Applied Remote Sensing Spring 2024
- GEOG/LDARCH C188: Geographic Information Systems Fall 2021

Seoul National University

Teaching Assistant

- 457.542*: Advanced Surveying (*Head TA*) Spring 2021
- 457.205: Introduction to Geospatial Engineering (*Lab Tutor & Head TA*) Spring 2021
- 457.539*: Advanced Remote Sensing: VHR Imagery (*Head TA*) Fall 2020
- 457.402: Remote Sensing (*Lab Tutor & Head TA*) Fall 2020
- 457.544*: Satellite Image Interpretation (*Head TA*) Spring 2020
- Leadership for Civil Engineers (*TA*) Spring 2020
- 457.205: Spatial Informatics and Systems (*Lab Tutor & Head TA*) Spring 2020

*Graduate-level Courses

Mentored Students at UC Berkeley

- Stella Wing (BS Conservation and Resource Studies & Data Science) **Sep 2023 – May 2024**
Current: MESM @ UCSB
- Harrison Raine (MLA & MCP) **Sep 2023 – July 2024**
Current: NASA
- Zeff Fengze Lin (Visiting student from South China University of Technology) **Jan 2023 – May 2023**
Current: Ph.D. @ Tsinghua U
- Weixin Li (MS Civil & Environmental Engineering) **Sep 2022 – May 2023**
Current: Ph.D. @ UCSB
- Xihan Yao (MLA) **Sep 2022 – May 2023**
Current: Ph.D. @ UT Austin
- Madison Chi (BS Environmental Science & Sustainable Design) **Sep 2022 – May 2023**
Current: MPH @ UCLA

Mentored Students at Seoul National University

- Hyounghoo Choi (BS Civil and Environmental Engineering) **Sep 2020 – Feb 2021**

SERVICES

Reviewer (Total: 49 reviews)

Agronomy (3), Applied sciences (2), European journal of remote sensing (1), Fire (2), Forecasting (3), Geospatial information science (2), Geocarto international (1), GIScience & remote sensing (3), IEEE JSTARS (2), International journal of Digital Earth (2), ISPRS international journal of geo-information (2), Land (2), Remote sensing (22), Sustainability (2)

Membership

- American Association of Geographers **2025**
- Association for Fire Ecology **2025**
- American Geophysical Union **2025**
- International Association of Wildland Fire **2025**
- Association of Environmental Professionals **2024**
- IEEE Geoscience and Remote Sensing Society **2023**
- American Society for Photogrammetry and Remote Sensing **2022**
- Korean-American Scientists and Engineers Association **2022**
- Korean Graduate Student Association **2021**
- International Society for Photogrammetry and Remote Sensing Student Consortium **2020**
- Korean Society of Civil Engineers **2019**

Student Committee, California Fire Science Consortium **2025**

Session Chair (Image Analysis for Land Cover Mapping), IGARSS 2023 **July 2023**

Geospatial Data Support, Investigative Reporting Program **Sep 2022 – Dec 2022**

- Developed and managed multi-modal geospatial data (multispectral satellite images, nighttime light images, vector data to map deforestation and human activity in Brazil using Google Earth Engine (geemap & Javascript) and QGIS for J298 OSINT Seminar in the School of Journalism at UC Berkeley.
- Provided workshops and advised journalists on GIS and geospatial tools.

Ammunition Inspector, Republic of Korea Army **May 2017 – Jan 2019**

- Recorded ammunition transactions and composed ammunition inventory reports. After working hours, contributed to write-up on pan-sharpening image fusion research using Worldview images [P1].

General Education Peer Tutor, Seoul National University **Mar 2016 – June 2016**

- Tutored college-level English to undergraduate students for incoming freshmen

Section Editor, The SNU Quill - SNU's English Press **Sep 2013 – June 2015**

- SNU campus news section reporter and editor for 9 volumes; responsible for 6-8 journal reporters. Also coordinated English writing/composition workshops and orientations.

PROJECTS

Roles:  GSR/RA |  Grant Writing |  Project Manager

Awarded Projects

 	Multi-scale mitigation of wildfire risk vulnerabilities in the natural and built environments (College of Environmental Design, UC Berkeley)	\$35,000	2025-2026
  	Deep Learning Framework for Mapping Basic Spatial Data in North Korea Using Very High Resolution Satellite Imagery (Institute for Peace and Unification)	\$28,000	2021-2022
 	Forecasting Photovoltaic Power using Meteorological Satellite Imagery and Deep Learning (SK Telecom)	\$60,000	2019

SKILLS

GitHub	https://github.com/minhokim93
Coding	Python, C/C++, CMake, Matlab, Javascript (Google Earth Engine), Git, Scripting (Bash), LaTeX, HTML
ML/DL	Tensorflow, Keras, PyTorch, Scikit-learn, Scikit-image, OpenCV
Remote Sensing	ENVI (SARscape), Google Earth Engine (Javascript/geemap), Python (Rasterio/GDAL)
GIS	ArcGIS, QGIS, Python
Languages	English (Native), Korean (Native), French (Fluent)

Certificates

- Teaching and Learning in Higher Education **2026**
- Geospatial Information Science and Technology **2022**

REFERENCES

Dr. Prof. Marta C. González

Department of Civil and Environmental Engineering
 Department of City and Regional Planning
 University of California, Berkeley
 406C Wurster Hall
 Email: martag@berkeley.edu

Dr. Prof. John Radke

Department of Landscape Architecture and Environmental Planning
 Department of City and Regional Planning
 University of California, Berkeley
 412 Wurster Hall #2000
 Email: ratt@berkeley.edu

Marc R. Castellnou

Incident Commander and Lead of Forest Actions Reinforcement Group (GRAF)
 Catalan Fire Service
 Avinguda de Serra Galliners, 112, 08290, Cerdanyola del Vallès, Spain
 Email: incendi@yahoo.com

Dr. Prof. Yongil Kim

Department of Civil and Environmental Engineering
 Seoul National University
 Building 35, Room 410 1, Gwanak-ro, Gwanak-gu, Seoul, 08826
 E-mail: yik@snu.ac.kr